UGC NET Environmental Science

UNIT 1: Fundamentals of Environmental Science

Complete Notes with Flowcharts, Tables & Memory Aids — 100% Score Guide

10 Topics | **100 MCQs** | **High Priority** | **8-10 Qs/Exam**

EXAM OVERVIEW — UNIT 1 AT A GLANCE

Parameter	Detail
Paper	UGC NET Paper 2 — Environmental Science (Subject Code 89)
Total questions from Unit 1	8–10 questions (out of 100 total)
Marks contribution	16–20 marks (out of 200 total)
Priority level	HIGH — Foundation for all other units
Most tested subtopics	Biogeochemical cycles, Energy flow, Ecosystem structure, Ecology terms
Question types	Direct factual, conceptual MCQs, numerical (energy calculations)
Suggested study time	5–6 days (first unit to study)

TABLE OF CONTENTS

1. SCOPE, DEFINITION & INTERDISCIPLINARY NATURE

Environmental Science is an **interdisciplinary field** that studies the interactions between humans and the natural world. It integrates knowledge from biology, chemistry, physics, geography, economics, sociology, and law to understand and solve environmental problems.

Key Term	Definition	Coined by / Year
Ecology	Study of relationships between organisms and their environment	Ernst Haeckel, 1866
Ecosystem	Functional unit of nature — biotic + abiotic interacting together	A.G. Tansley, 1935
Environment	Sum of all external conditions affecting life of an organism	General usage
Biotic factors	Living components: plants, animals, microbes, fungi	—
Abiotic factors	Non-living components: temperature, light, water, soil, air, pH	—
Autecology	Study of a single species / organism with its environment	—
Synecology	Study of a group of organisms (community) with their environment	—
Biosphere	Global zone of life; overlap of atmosphere, hydrosphere, lithosphere	Eduard Suess, 1875

EXAM HOTSPOT

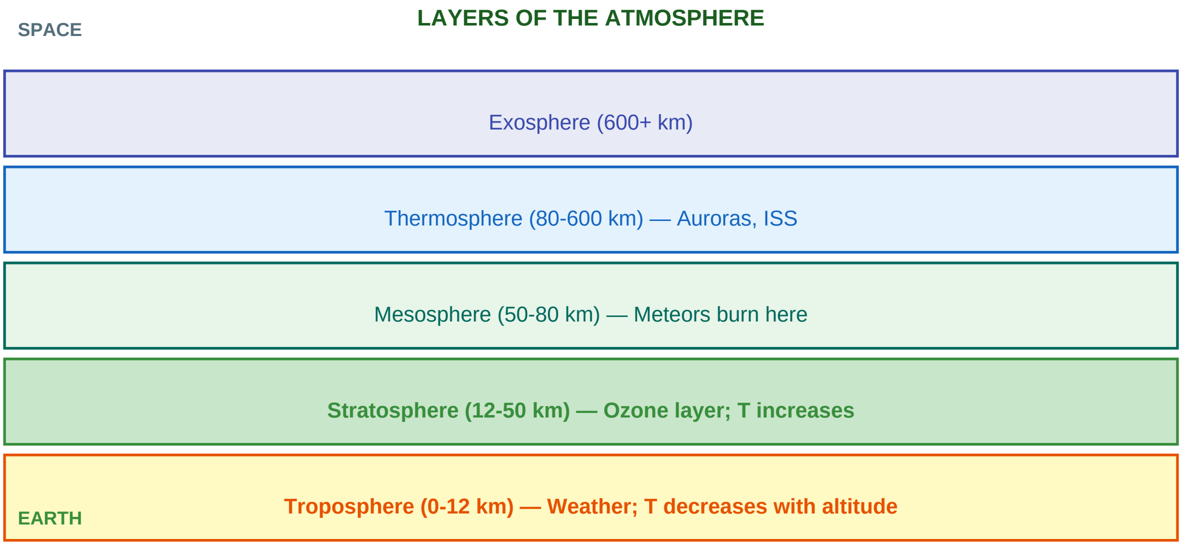
Tansley coined 'ECOSYSTEM' (1935). Haeckel coined 'ECOLOGY' (1866). These two are directly asked in almost every UGC NET exam. Do NOT confuse them.

Interdisciplinary nature of Environmental Science: Biology (ecology, biodiversity), Chemistry (pollution, cycles), Physics (energy flow, radiation), Geography (biomes, climate), Economics (resource valuation), Sociology (human behaviour, policy), Law (environmental regulations).

2. STRUCTURE OF EARTH'S SYSTEMS — THE 4 SPHERES

Sphere	Definition	Components	Depth/Range
Atmosphere	Gaseous envelope surrounding Earth	Troposphere, Stratosphere, Mesosphere, Thermosphere, Exosphere	Up to ~10,000 km
Hydrosphere	All water on/near Earth's surface	Oceans (97%), glaciers, rivers, lakes, groundwater, atmospheric water	Surface to deep ocean
Lithosphere	Outer solid rocky shell of Earth	Crust (continental + oceanic) + upper mantle; rocks, soil, minerals	0–100 km depth
Biosphere	Zone where life exists on Earth	Overlap zone of atmosphere+hydrosphere+lithosphere; all living organisms	~10 km above – 10 km below sea level

2.1 Layers of the Atmosphere — Detailed



Layer	Altitude	Temperature trend	Key features
Troposphere	0–12 km	Decreases (lapse rate: 6.5°C/km)	ALL weather occurs here; 75% of atmosphere mass; most clouds
Tropopause	~12 km	Constant (–56°C)	Boundary — temperature stops decreasing; acts as 'lid'
Stratosphere	12–50 km	INCREASES with altitude (inversion)	OZONE LAYER (20–35 km); calm, cloudless; used by aircraft
Stratopause	~50 km	Maximum (~0°C)	Boundary between stratosphere and mesosphere
Mesosphere	50–80 km	Decreases (coldest layer: –90°C)	METEORS burn here due to friction; noctilucent clouds
Thermosphere	80–600 km	Increases rapidly	AURORAS (Northern/Southern lights); International Space Station
Exosphere	600+ km	Very high but diffuse	Outermost layer; merges with space; satellites orbit here

MEMORY AID — Atmosphere Layers

TSMT-E = Troposphere, Stratosphere, Mesosphere, Thermosphere, Exosphere Trick: 'The Sky Makes Tall Elephants' Temperatures: DOWN in Troposphere, UP in Stratosphere, DOWN in Mesosphere, UP in Thermosphere

3. ECOSYSTEM — CONCEPT, STRUCTURE & FUNCTION

Definition: An ecosystem is a functional, self-sustaining unit of nature where living organisms (biotic) interact with each other and with their non-living (abiotic) environment, involving energy flow and matter cycling. Term coined by A.G. Tansley (1935).

ECOSYSTEM

BIOTIC COMPONENTS

- Producers (Autotrophs)
- Primary Consumers
- Secondary Consumers
- Tertiary Consumers
- Decomposers / Detritivores

ABIOTIC COMPONENTS

- Temperature / Light / Wind
- Water / Humidity
- Soil / Minerals / pH
- Atmospheric gases (O■, CO■)
- Inorganic nutrients (N, P, S)

3.1 Biotic Components — Detailed

Component	Also called	Role	Examples
Producers	Autotrophs / 1st trophic level	Fix solar energy via photosynthesis; produce organic matter from inorganic	Green plants, phytoplankton, cyanobacteria, chemosynthetic bacteria
Primary consumers	Herbivores / 2nd trophic level	Feed on producers; first level of energy transfer	Grasshoppers, deer, cows, rabbits, zooplankton
Secondary consumers	Carnivores-1 / 3rd trophic level	Feed on herbivores	Frogs, small fish, foxes, snakes
Tertiary consumers	Carnivores-2 / 4th trophic level	Feed on secondary consumers (top predators)	Hawks, lions, sharks, large eagles
Decomposers	Saprotrophic microbes	Break down dead organic matter into inorganic nutrients; recycle nutrients	Bacteria (Bacillus), Fungi (Aspergillus, Rhizopus)
Detritivores	Macro-decomposers	Feed on detritus (partially decomposed organic matter)	Earthworms, millipedes, wood lice, some beetles

3.2 Types of Ecosystems

Type	Sub-types	Key feature	NPP ($g/m^2/yr$)
Terrestrial	Forests (tropical, temperate, boreal), Grasslands, Deserts, Tundra	Climate determines biome type	140 (tundra) to 2200 (tropical rainforest)
Aquatic — Freshwater	Lakes, ponds, rivers, streams, wetlands	Low salinity (<0.5 ppt); high biodiversity per area	200–1500
Aquatic — Marine	Open ocean, coral reefs, estuaries, mangroves	High salinity (~35 ppt); largest ecosystem by area	125 (open ocean) to 2500 (estuaries)
Artificial	Agro-ecosystems, urban ecosystems, aquaculture	Human-managed; low species diversity; high productivity	Varies

EXAM FACT

Tropical rainforests have HIGHEST NPP on land (~2000-2200 $g/m^2/yr$). Open oceans have LOWEST NPP per unit area (~125 $g/m^2/yr$) but are largest by area. Estuaries + coral reefs have HIGHEST marine NPP. Deserts and tundra have LOWEST NPP.

4. ECOSYSTEM SERVICES — 4 CATEGORIES (Millennium Ecosystem Assessment)

Category	Definition	Examples	Exam weight
Provisioning services	Direct products obtained FROM ecosystems	Food, freshwater, timber, fibre, medicines, genetic resources, fuel	HIGH
Regulating services	Benefits from REGULATION of ecosystem processes	Climate regulation, carbon sequestration, water purification, flood control, disease regulation, pollination	HIGH
Cultural services	NON-MATERIAL benefits — psychological, spiritual, recreational	Recreation, ecotourism, spiritual values, aesthetic beauty, education, cultural heritage	MEDIUM
Supporting services	Services that SUPPORT all other services (indirect)	Soil formation, nutrient cycling, primary production, oxygen production, water cycling	MEDIUM

TRICK TO REMEMBER

PRCS = Provisioning, Regulating, Cultural, Supporting Memory: 'Plants Regulate Culture and Support life' KEY TRAP: Carbon sequestration = REGULATING (not supporting). Soil formation = SUPPORTING. Food = PROVISIONING.

5. ENERGY FLOW — LAWS OF THERMODYNAMICS & 10% LAW

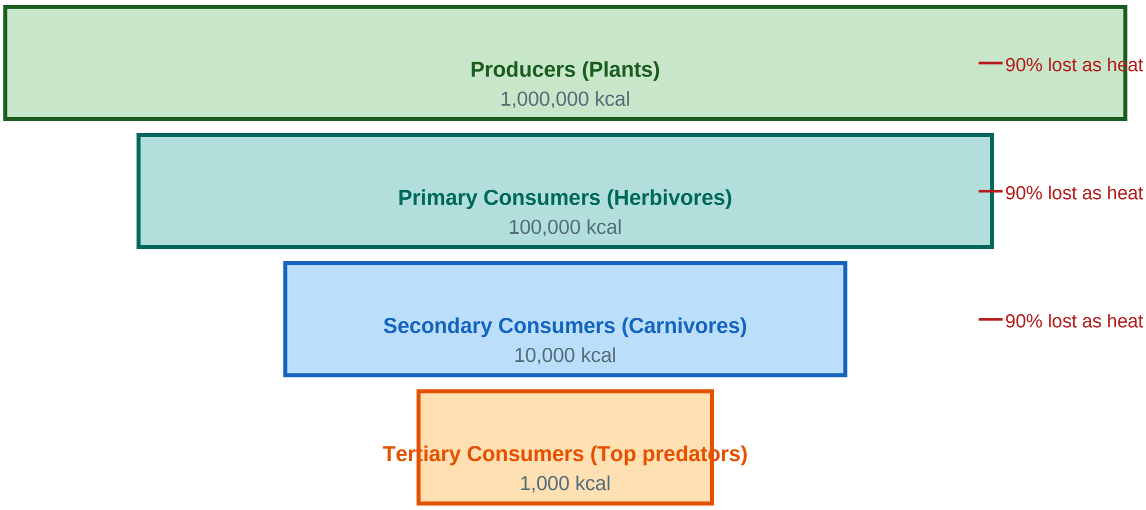
5.1 Laws of Thermodynamics in Ecology

Law	Statement	Ecological implication
First Law (Conservation)	Energy can neither be created nor destroyed; only converted from one form to another	Total energy entering an ecosystem = energy stored + energy lost; energy is conserved
Second Law (Entropy)	Every energy conversion increases entropy; no conversion is 100% efficient; energy degrades to heat	Energy flow is UNIDIRECTIONAL and DECREASING — cannot be recycled like matter; food chains are short

5.2 Lindeman's 10% Law (1942) — Energy Transfer

Only 10% of energy at one trophic level transfers to the next higher level. 90% is lost as: heat (respiration), excretion, incomplete digestion, and dead organic matter. This is why food chains are limited to 4–5 trophic levels — too little energy remains at higher levels.

LINDEMAN'S 10% LAW — Energy Flow Pyramid



Trophic Level	Example	Energy (starting from 1,000,000 kcal)	% of original
1st — Producers	Grass / Phytoplankton	1,000,000 kcal	100%
2nd — Primary consumers	Grasshopper / Zooplankton	100,000 kcal	10%
3rd — Secondary consumers	Frog / Small fish	10,000 kcal	1%
4th — Tertiary consumers	Snake / Large fish	1,000 kcal	0.1%
5th — Quaternary consumers	Hawk / Shark	100 kcal	0.01%

5.3 Types of Productivity

Term	Formula/Definition	What it means
Gross Primary Productivity (GPP)	Total photosynthesis rate	ALL energy fixed by producers (including what they use for respiration)
Autotrophic Respiration (Ra)	Respiration by producers	Energy used by plants for their own metabolism
Net Primary Productivity (NPP)	NPP = GPP – Ra	Energy available to consumers and decomposers; what ecologists usually measure
Net Ecosystem Production (NEP)	NEP = NPP – Rh	Rh = heterotrophic respiration. NEP = net C gain/loss of ecosystem
Secondary Productivity	Energy stored by consumers	Rate of biomass accumulation by heterotrophs (consumers, decomposers)

KEY EXAM POINTS on Energy

1. Energy flow = UNIDIRECTIONAL (one way: sun to producers to consumers). Matter/nutrients = CYCLIC. 2. Shortest food chain = maximum energy for top consumer. Veganism is more energy-efficient than meat-eating. 3. Maximum growth rate in logistic growth occurs at N = K/2 (not at K, not at 0).

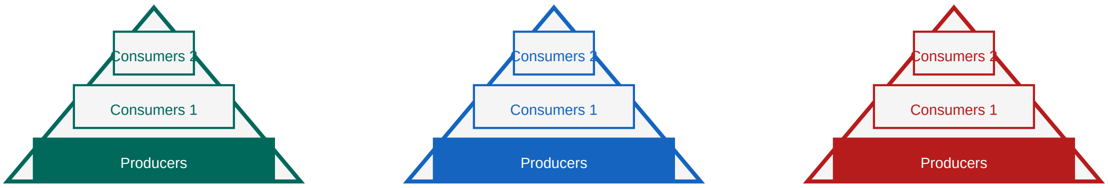
5.4 Types of Food Chains

Type	Starts with	Example chain	Where dominant
Grazing Food Chain (GFC)	Living green plants (producers)	Grass → Grasshopper → Frog → Snake → Hawk	Grasslands, tropical forests, aquatic (upper layers)
Detritus Food Chain (DFC)	Dead organic matter (detritus)	Dead leaves → Earthworm → Robin → Sparrowhawk	Forest floor, soil, deep ocean, decomposition-rich areas

Food Web: A network of interconnected food chains in an ecosystem. More realistic than a single food chain. Provides STABILITY — if one species disappears, energy can flow through alternative paths. Greater food web complexity = greater ecosystem resilience.

6. ECOLOGICAL PYRAMIDS — ALL 3 TYPES

THREE ECOLOGICAL PYRAMIDS — KEY DIFFERENCES



Pyramid of Numbers
Can be inverted
(tree ecosystem)

Pyramid of Biomass
Can be inverted
(aquatic ecosystem)

Pyramid of Energy
ALWAYS UPRIGHT
(NEVER inverted)

Pyramid	What it shows	Can it be inverted?	When inverted?	Proposed by
Pyramid of Numbers	Number of organisms at each trophic level	YES	Tree ecosystem (1 tree → many insects → fewer birds) OR parasitic chain	Charles Elton (1927)
Pyramid of Biomass	Total dry weight (biomass) at each trophic level	YES	Aquatic ecosystems (phytoplankton has low standing biomass but high turnover; zooplankton may outweigh it)	Charles Elton
Pyramid of Energy	Energy content at each trophic level per unit area per unit time	NEVER (ALWAYS UPRIGHT)	Cannot be inverted — energy always decreases upward (2nd law of thermodynamics)	H.T. Odum (1957)

CRITICAL EXAM FACT — Most Frequently Asked

Pyramid of ENERGY is ALWAYS UPRIGHT. This is the most repeated question in UGC NET EVS. Reason: Second law of thermodynamics ensures energy always decreases at higher trophic levels. Numbers and biomass can be inverted; energy NEVER can be.

7. BIOGEOCHEMICAL CYCLES — COMPLETE COVERAGE

BIOGEOCHEMICAL CYCLES — Quick Comparison

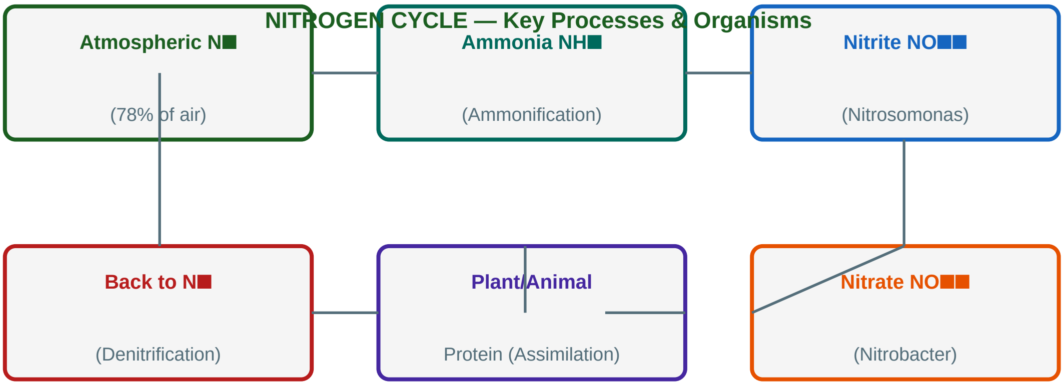
	Carbon (C)	Nitrogen (N)	Phosphorus (P)	Sulphur (S)
Gaseous phase?	Yes (CO■, CH■)	Yes (N■, NH■, N■O)	NO (sedimentary only)	Yes (SO■, H■S)
Key reservoir	Oceans/atmosphere	Atmosphere (78%)	Rocks/sediments	Rocks/oceans
Key process	Photosynthesis/ Respiration	N■-fixation/ Denitrification	Weathering/ Assimilation	Oxidation/ Reduction
Key organisms	Plants, decomposers	Rhizobium, Nitrobacter	Decomposers, plants	Thiobacillus
Exam weight	HIGH	HIGHEST	HIGH	MEDIUM

7.1 Carbon Cycle

Process	Direction	Key organisms/agents	Approx rate
Photosynthesis	Atmosphere → Biosphere (CO2 in)	Green plants, phytoplankton, cyanobacteria	120 Gt C/yr (terrestrial)
Respiration	Biosphere → Atmosphere (CO2 out)	All living organisms	~60 Gt C/yr
Decomposition	Biosphere → Soil/Atmosphere	Bacteria, fungi (decomposers)	Slow; years to centuries
Combustion	Lithosphere/Biosphere → Atmosphere	Fossil fuel burning, wildfires	~10 Gt C/yr (anthropogenic)
Ocean absorption	Atmosphere → Hydrosphere	Physical dissolution, marine photosynthesis	~2.5 Gt C/yr
Volcanic activity	Lithosphere → Atmosphere (CO2 out)	Geological process	~0.2 Gt C/yr

Carbon Reservoir	Amount (Gt C)	Type
Oceans (dissolved)	~38,000	Largest reservoir; acts as carbon sink
Fossil fuels + sediments	~10,000	Long-term geological store
Soil organic matter	~2,000	Important terrestrial sink
Atmosphere	~860	CO2 ~420 ppm currently (pre-industrial: 280 ppm)
Terrestrial vegetation	~560	Forests are major carbon sinks

7.2 Nitrogen Cycle — MOST IMPORTANT CYCLE



Process	Conversion	Key organisms	Conditions
Nitrogen fixation	N2 (gas) → NH3 (ammonia)	Rhizobium (symbiotic in legumes), Azotobacter (free-living aerobic), Clostridium (free-living anaerobic), Cyanobacteria (aquatic)	High energy requirement; enzyme nitrogenase
Ammonification	Organic N → NH3/NH4+	Decomposers (bacteria, fungi) — e.g., Bacillus, Pseudomonas	Decomposition of dead organic matter
Nitrification Step 1	NH4+ → NO2-	Nitrosomonas bacteria	Aerobic conditions; soil
Nitrification Step 2	NO2- → NO3-	Nitrobacter bacteria	Aerobic conditions; soil
Assimilation	NO3-/NH4+ → Organic N (protein)	Plant roots absorb nitrates	Active uptake by plant roots
Denitrification	NO3- → N2 (gas) released to atmosphere	Pseudomonas, Thiobacillus, Paracoccus	ANAEROBIC conditions (waterlogged soils)

NITROGEN CYCLE EXAM TIPS — Very Frequently Asked

1. Nitrification = NH4+ → NO2- (Nitrosomonas) → NO3- (Nitrobacter). Both are AEROBIC. 2. Denitrification = NO3- → N2. Done by ANAEROBIC bacteria (Pseudomonas) in waterlogged soils. 3. Rhizobium = symbiotic (in legume nodules). Azotobacter = free-living aerobic. Clostridium = free-living anaerobic. 4. Lightning also fixes small amounts of N2 → NO (industrial fixation analogy).

7.3 Phosphorus Cycle

Feature	Detail
Type	SEDIMENTARY cycle — NO gaseous phase (key distinguishing feature)
Major reservoir	Phosphate rocks and marine sediments
Key pathway	Rocks → Weathering → Soil → Plant roots (absorb H2PO4-) → Animals → Decomposers → Soil → Oceans → Sediments → Rocks (geological uplift)
Limiting nutrient	Often the LIMITING NUTRIENT in aquatic ecosystems (not nitrogen)
Environmental problem	Excess P from fertilisers and detergents → EUTROPHICATION of water bodies
No gaseous phase	Cannot enter atmosphere significantly — this makes P cycle very slow and easily disrupted

7.4 Water (Hydrological) Cycle

Process	Definition	Key data
Evaporation	Water from oceans, lakes, rivers converts to water vapour	~502,000 km3/yr from oceans
Transpiration	Water loss from plant leaves as vapour	Forests transpire significant amounts
Evapotranspiration	Combined evaporation + transpiration from land surfaces	Major pathway of water from land to atmosphere
Condensation	Water vapour cools and forms clouds/mist	Occurs when air reaches dew point
Precipitation	Water returns to Earth as rain, snow, sleet, hail	~111,000 km3/yr on land
Infiltration	Water soaks into soil and recharges groundwater	Rate depends on soil texture
Runoff / Surface flow	Water flows over land surface to rivers and oceans	Increases with deforestation, urbanisation

Water Reservoir	% of total water	Significance
Oceans	~97.5%	Largest reservoir; salt water; major evaporation source
Glaciers / Ice caps	~1.74%	Largest FRESHWATER reservoir (~69% of freshwater)
Groundwater	~0.76%	~30% of all freshwater; important drinking water source
Surface freshwater	~0.013%	Rivers, lakes — easily accessible but tiny fraction
Atmosphere	~0.001%	Small but critical for weather patterns

7.5 Sulphur Cycle

Process	Conversion	Key organisms/agents
Weathering	Sulphate minerals in rocks → dissolved SO4(2-)	Geological weathering processes
Biological assimilation	SO4(2-) → Organic S (in proteins, amino acids)	Plants, microorganisms
Decomposition/Ammonification	Organic S → H2S (hydrogen sulphide)	Desulfovibrio (anaerobic bacteria)
Oxidation	H2S → SO2 → SO4(2-)	Thiobacillus (aerobic bacteria); atmospheric oxidation
Volcanic activity	S compounds → SO2 released	Major natural source of atmospheric SO2
Fossil fuel combustion	Organic S in coal/oil → SO2 (anthropogenic)	Major cause of acid rain
Acid rain formation	SO2 + OH radicals → H2SO4 (sulfuric acid)	Atmospheric chemistry; damages buildings, forests, lakes

8. PHYSICO-CHEMICAL (ABIOTIC) FACTORS & ORGANISM RESPONSES

Factor	Key concepts	Organism classification / Response	Exam importance
Temperature	Most important abiotic factor. Governs metabolic rates. Poikilotherms (cold-blooded): body T = ambient. Homeotherms (warm-blooded): maintain constant T.	Stenotherm: narrow T range (coral, polar bear). Eurytherm: wide T range (humans, cockroach, fish). Q10 rule: metabolic rate doubles per 10°C increase.	HIGH
Light	Drives photosynthesis. PAR (Photosynthetically Active Radiation): 400-700 nm. Photoperiodism: day length triggers flowering, migration, reproduction.	Photophilous: sun-loving. Sciophilous (shade-tolerant): forest floor plants. Compensation point: photosynthesis = respiration (no net CO_2 exchange).	HIGH
Water/Humidity	Essential for all life. Availability shapes biome distribution.	Xerophytes: dry environments (cactus, acacia). Mesophytes: moderate water. Hydrophytes: aquatic (lotus, water lily). Hygrophytes: high humidity forests.	MEDIUM
Soil (Edaphic factors)	Texture (clay, silt, sand), structure, pH, organic matter content (humus), CEC (Cation Exchange Capacity).	Calcicoles: prefer alkaline/calcareous soils. Calcifuges: prefer acidic soils. Optimal pH 6-7 for most crops. Acidic soils: Al/Mn toxicity. Alkaline: Fe/Zn deficiency.	MEDIUM
Wind	Aids dispersal of seeds/pollen. Increases transpiration. Shapes vegetation (Krummholz at treeline). Mechanical stress on trees.	Anemophily: wind pollination (grasses, conifers). Anemochory: wind seed dispersal. Krummholz: stunted trees at alpine treeline due to wind and snow.	LOW-MEDIUM
pH	Controls nutrient availability, microbial activity, chemical weathering.	Acidic soils: conifers, heather, rhododendron. Neutral: most agricultural crops. Alkaline: cacti, many desert plants.	MEDIUM

Liebig's Law vs Shelford's Law — Critical Distinction

LIEBIG'S LAW OF MINIMUM (1840): Growth is controlled by the MOST LIMITING (scarcest) essential resource — not by total resources available. Like a barrel leaking at the shortest stave. SHELFORD'S LAW OF TOLERANCE (1913): Each organism has a MINIMUM, OPTIMUM, and MAXIMUM for each environmental factor. Performance peaks at optimum; organisms die beyond minimum/maximum limits.

9. ECOLOGICAL CONCEPTS — NICHE, HABITAT, SUCCESSION & MORE

9.1 Habitat vs Ecological Niche

Concept	Definition	Analogy	Example
Habitat	The physical place/location where an organism lives	ADDRESS of the organism	A pond is the habitat of a frog
Ecological Niche	The role/function of an organism in its ecosystem — what it eats, when it is active, what it eats, how it interacts with other species	PROFESSION/JOB of the organism	The frog's niche: eats insects, active at dusk, controls insect populations, prey for snakes
Fundamental Niche	Theoretical maximum range of conditions an organism can use (without competition)	Potential niche	Hutchinson (1957): n-dimensional hypervolume
Realized Niche	Actual niche occupied in presence of competition and predation	Actual niche (always ≤ fundamental)	Reduced by competitive interactions

Competitive Exclusion Principle (Gause's Law, 1934): Two species competing for exactly the same ecological niche CANNOT coexist indefinitely in the same habitat. One will outcompete the other and drive it to local extinction OR one species will evolve to occupy a slightly different niche (character displacement / niche partitioning).

9.2 Ecological Succession

PRIMARY ECOLOGICAL SUCCESSION

Bare rock / Barren land → Pioneer species (Lichens, Mosses) → Intermediate seral stages (shrubs) → Climax Community (Forest)

Feature	Primary Succession	Secondary Succession
Starting substrate	BARE rock / sand / new volcanic island — NO soil, no organisms	Disturbed area WITH soil and some organisms remaining
Pioneer species	Lichens (most important — break down rock), then mosses, then grasses	Grasses, weeds, fast-growing annuals
Duration	Very slow — hundreds to thousands of years	Faster — decades to centuries
Examples	New volcanic island (Hawaii), exposed glacial till, new sandbar	Abandoned farmland, burned forest, flooded land after drainage
Climax community	Reaches regional climax (determined by climate)	Returns to pre-disturbance community (if enough time)
Soil development	Must develop from scratch via rock weathering	Soil already present; just recovering

Succession Trend	Direction of Change
Species diversity	INCREASES throughout succession
Biomass	INCREASES (more organic matter accumulates)
Productivity (NPP)	Increases then stabilises at climax

Succession Trend	Direction of Change
Complexity/food web	INCREASES (more species, more interactions)
Stability	INCREASES toward climax
Nutrient retention	INCREASES (less leaching)
Respiration:Photosynthesis ratio	Moves toward 1 (equilibrium) at climax

9.3 Population Ecology

Concept	Formula/Details	Exam relevance
Exponential growth	dN/dt = rN; J-shaped curve; unlimited resources; r = intrinsic rate of natural increase	Describes rapid population growth when resources are unlimited (bacteria, introduced species)
Logistic growth	dN/dt = rN(K-N)/K; S-shaped (sigmoidal) curve; K = carrying capacity	More realistic; used for most natural populations
Carrying capacity (K)	Maximum population size the environment can sustain indefinitely	Population stabilizes at K in logistic growth
Maximum growth rate	Occurs when N = K/2 (half the carrying capacity)	VERY FREQUENTLY ASKED — not at K, not at 0
r-selected species	High r (reproductive rate), many small offspring, short lifespan, Type III survivorship	Examples: bacteria, insects, annual plants, mice
K-selected species	Low r, few large offspring, long lifespan, high parental care, Type I survivorship	Examples: elephants, whales, humans, oak trees

9.4 Species Interactions

Interaction	Effect on A / B	Definition	Examples
Mutualism	+/+	BOTH species benefit	Mycorrhizae (fungi+plants), Rhizobium+legumes, clownfish+anemone, lichens, pollination
Commensalism	+/0	One benefits, other UNAFFECTED	Epiphytic orchids on tree, barnacles on whale, egrets following cattle, remora fish on shark
Amensalism	-/0	One is HARMED, other unaffected	Allelopathy: black walnut releases juglone killing nearby plants; penicillin inhibiting bacteria
Predation	+/-	Predator benefits; prey harmed	Lion-zebra, spider-fly, hawk-mouse, lynx-hare (classic predator-prey oscillation)
Parasitism	+/-	Parasite benefits; host harmed but usually not killed immediately	Tapeworm in human intestine, Plasmodium (malaria), mistletoe on tree, cuckoo (brood parasitism)
Competition	-/-	BOTH species harmed (compete for same resource)	Intraspecific: same species. Interspecific: different species. Can lead to competitive exclusion.

10. BIODIVERSITY — TYPES, HOTSPOTS & CONSERVATION

Type of Diversity	Definition	Scale	Measured by
Alpha (α) diversity	Species richness AND evenness WITHIN a single community/habitat	Local/site level	Species richness, Shannon-Wiener index, Simpson's index
Beta (β) diversity	CHANGE/TURNOVER in species composition BETWEEN different habitats	Between habitats	Whittaker's beta = gamma/alpha; Jaccard and Sorensen similarity indices
Gamma (γ) diversity	Total diversity across a LANDSCAPE or region (all habitats combined)	Landscape/regional level	Total species count across all habitats in a region

Biodiversity Hotspot	Definition	Criteria (Conservation International)
Biodiversity Hotspot	Area with exceptionally high endemism under severe threat of destruction	≥1,500 endemic vascular plant species AND has lost ≥70% of original native habitat
Concept origin	Norman Myers (1988) — originally 10 hotspots; now 36 globally	36 hotspots cover only ~2.5% of Earth's land but harbour >50% of all endemic plant species

India's 4 Biodiversity Hotspots	Location	Key features
Western Ghats + Sri Lanka	Western coast of India + Sri Lanka	High rainfall, rich in endemic amphibians and plants; under severe agricultural threat
Himalaya (Eastern Himalaya)	Northeast India, Nepal, Bhutan, southern China	Highest mountain range; diverse flora from tropical to alpine; snow leopard habitat
Indo-Burma	Northeast India, Myanmar, Thailand, Cambodia	Extremely high endemism in freshwater turtles, plants, mammals
Sundaland (partial)	Parts of Andaman & Nicobar Islands	Island biodiversity; endemic to island arcs of Southeast Asia

Conservation Type	Methods	Examples
In-situ (in natural habitat)	National parks, wildlife sanctuaries, biosphere reserves, sacred groves, community reserves, Ramsar wetlands	Jim Corbett NP, Kaziranga, Sundarbans Biosphere Reserve, Chilika Ramsar Site
Ex-situ (outside natural habitat)	Zoos, botanical gardens, aquaria, gene banks, seed banks, cryopreservation, captive breeding	National Bureau of Plant Genetic Resources (NBPGR), Svalbard Global Seed Vault

IUCN Red List Category	Criteria	Example species
Extinct (EX)	No reasonable doubt that last individual has died	Dodo, Passenger pigeon, Thylacine
Extinct in Wild (EW)	Survives only in captivity/cultivation	Przewalski's horse (recovered), Scimitar-horned oryx
Critically Endangered (CR)	Extremely high risk of extinction in wild	Amur leopard, Javan rhino, Sumatran orangutan
Endangered (EN)	Very high risk of extinction	Blue whale, African wild dog, Snow leopard
Vulnerable (VU)	High risk of extinction	Hippopotamus, Polar bear, Whale shark
Near Threatened (NT)	Close to qualifying as threatened	Jaguar, White shark
Least Concern (LC)	Not at significant risk	House sparrow, Brown rat, Humans

11. QUICK REFERENCE — SCIENTISTS, LAWS & KEY TERMS

Scientist	Year	Contribution	Exam frequency
Ernst Haeckel	1866	Coined the term 'Ecology' (Oekologie)	VERY HIGH
A.G. Tansley	1935	Coined the term 'Ecosystem'	VERY HIGH
Raymond Lindeman	1942	10% Law of energy transfer (trophic efficiency)	VERY HIGH
G.E. Hutchinson	1957	Niche concept — n-dimensional hypervolume; fundamental vs realized niche	HIGH
G.F. Gause	1934	Competitive Exclusion Principle	HIGH
E.P. Odum	1953	Fundamentals of Ecology textbook; popularised ecosystem ecology	HIGH
Charles Elton	1927	Food chain, food web, ecological pyramid concepts; 'Animal Ecology' book	HIGH
H.T. Odum	1957	Quantified pyramid of energy in Silver Springs, Florida	MEDIUM
J. Liebig	1840	Law of Minimum — growth limited by scarcest nutrient	HIGH
V.E. Shelford	1913	Law of Tolerance — min, optimum, max for environmental factors	HIGH
James Lovelock	1970s	Gaia hypothesis — Earth as self-regulating system	MEDIUM
Norman Myers	1988	Biodiversity Hotspot concept — originally 10 hotspots	MEDIUM
Eduard Suess	1875	Coined 'Biosphere'	LOW
Charles Darwin	1859	Natural selection, evolution (ecological foundation)	LOW

Important Numerical / Data	Value
N2 in atmosphere	~78%
O2 in atmosphere	~21%
CO2 in atmosphere (current)	~420 ppm (pre-industrial: 280 ppm; increase ~50%)
Troposphere thickness	0–12 km (varies: ~8 km at poles, ~16 km at equator)
Stratosphere ozone layer	20–35 km altitude
Lapse rate in troposphere	6.5°C per km (temperature decreases with altitude)
Tropopause temperature	~–56°C
Oceans = % of Earth's water	~97.5% (saltwater)
Glaciers = % of freshwater	~69%
Tropical rainforest NPP	~2000–2200 g/m2/yr (highest terrestrial)
Open ocean NPP	~125 g/m2/yr (lowest marine per unit area)
Tundra/desert NPP	~140 g/m2/yr (lowest terrestrial)
Energy transfer efficiency (10% law)	10% per trophic level (90% lost as heat)
No. of trophic levels typical	4–5 (limited by energy loss)
Biodiversity hotspot criteria: endemic plants	≥1,500 endemic vascular plant species
Biodiversity hotspot criteria: habitat loss	≥70% of original native habitat lost

Important Numerical / Data	Value
India's biodiversity hotspots	4 (Western Ghats+SL, Himalayas, Indo-Burma, Sundaland)

12. MNEMONICS, MEMORY TREES & EXAM STRATEGY

12.1 Master Mnemonic Sheet

Topic	Mnemonic / Memory Aid
Atmosphere layers (bottom to top)	'The Sky Makes Tall Elephants' = Troposphere, Stratosphere, Mesosphere, Thermosphere, Exosphere
Ecosystem services (4 types)	PRCS = Provisioning, Regulating, Cultural, Supporting | 'Plants Regulate Culture and Support life'
Nitrogen cycle steps	Fix-Am-Ni-Ni-As-Den = N2-fixation → Ammonification → Nitrification(1) → Nitrification(2) → Assimilation → Denitrification
Nitrification bacteria	NitroSOMOnas = NH_4^+ → NO_2^- (step 1). NitroBACTer = NO_2^- → NO_3^- (step 2). Remember: 'S before B; 2 before 3'
N2 fixers (free-living)	AC = Azotobacter (aerobic), Clostridium (anaerobic). Rhizobium = symbiotic in legumes
Pyramid of energy	ENERGY = ERECT always. Never inverted. 2nd law = one-way street.
Ecological interactions signs	Mutualism=++, Commensalism=+0, Amensalism=-0, Predation=+-, Parasitism=+-, Competition=--
Gaia hypothesis scientist	LOVE-LOCK loves the planet = James LOVELOCK
Ecology (1866) vs Ecosystem (1935)	HECK it's Haeckel (1866). TANSLEY coined ecoSYSTEM (1935). 'H before T; 1866 before 1935'
r vs K selection	r = Rabbit (many babies, fast). K = Killer whale (few babies, slow, big brain, long life)
IUCN Red List order	EX-EW-CR-EN-VU-NT-LC: 'Every Elephant Can Eat Very Nice Leaves Carefully'
Carbon reservoirs (largest to smallest)	Oceans > Fossil fuels/sediments > Soil > Atmosphere > Vegetation
India's 4 hotspots	WISH = Western Ghats, Indo-Burma, Sundaland (Andaman), Himalayas
In-situ vs Ex-situ	In-situ = IN natural habitat (National parks). Ex-situ = EXited from habitat (Zoo, Seed bank)
10% energy law calculation	Each step DIVIDES by 10. Start: 1,000,000 → 100,000 → 10,000 → 1,000 → 100
NPP formula	NPP = GPP – Ra. Remember: 'Producers must Pay Rent (Ra) from their Gross income; Net is what remains'

12.2 Most Repeated Topics in PYQs (2014–2023)

Rank	Topic	Avg. Qs/Exam	Type of question
1	Lindeman's 10% Law + energy calculation	1–2	Concept + numerical
2	Pyramid of energy always upright (never inverted)	1	Direct factual
3	Nitrogen cycle steps + organisms (Nitrosomonas, Nitrobacter, Rhizobium)	1–2	Organism identification
4	Term coined: Ecology (Haeckel), Ecosystem (Tansley)	1	Direct factual
5	Ecosystem services classification (PRCS)	1	Identification / matching
6	Phosphorus cycle = sedimentary (no gaseous phase)	1	Direct factual
7	Competitive exclusion principle (Gause's law)	1	Concept

Rank	Topic	Avg. Qs/Exam	Type of question
8	Fundamental vs realized niche (Hutchinson)	1	Concept + application
9	Primary vs secondary succession	1	Comparison / identification
10	Carbon reservoirs: oceans = largest	1	Direct factual
11	In-situ vs ex-situ conservation methods	1	Classification
12	IUCN Red List categories + India's hotspots	1	Factual

FINAL EXAM STRATEGY FOR UNIT 1

1. NEVER confuse Haeckel (ecology, 1866) with Tansley (ecosystem, 1935). 2. Pyramid of ENERGY is ALWAYS upright — repeat this 10 times until automatic. 3. Nitrogen cycle organism sequence: Rhizobium (fix) → Nitrosomonas (step 1 nitrification) → Nitrobacter (step 2). 4. Phosphorus = sedimentary = NO gaseous phase. This is asked almost every year. 5. For energy calculations: each step x 0.1 (divide by 10). 6. Maximum logistic growth = at N = K/2 (not K, not 0). 7. NPP = GPP – Ra. NEP = NPP – Rh. Know both. 8. Ecotone = transition zone with HIGHER biodiversity (edge effect). 9. India's 4 biodiversity hotspots: WISH = Western Ghats, Indo-Burma, Sundaland (Andaman), Himalayas. 10. Oceans = largest carbon AND water reservoir. Glaciers = largest FRESHWATER reservoir.

12.3 Common Exam Traps — Do NOT Fall For These

Common mistake	Correct answer
Pyramid of NUMBERS can be inverted → therefore all pyramids can be inverted	WRONG. Only numbers & biomass can invert. Energy pyramid NEVER inverts.
The largest freshwater reservoir is rivers/lakes (visible freshwater)	WRONG. Glaciers/ice caps hold ~69% of freshwater. Rivers+lakes = only 0.3% of freshwater.
Decomposers and detritivores are the same thing	WRONG. Decomposers = bacteria+fungi (microscopic, chemical breakdown). Detritivores = macroorganisms (earthworms, millipedes) that physically break up detritus.
Fundamental niche = realized niche in competition	WRONG. Realized niche is always SMALLER than or equal to fundamental niche due to competition.
Carbon sequestration is a SUPPORTING service	WRONG. Carbon sequestration = REGULATING service. Soil formation = SUPPORTING.
Azotobacter is symbiotic like Rhizobium	WRONG. Azotobacter is FREE-LIVING (aerobic). Rhizobium is SYMBIOTIC in legume root nodules.
Nitrification: Nitrobacter converts NH_4^+ to NO_2^-	WRONG. Nitrosomonas does $NH_4^+ \to NO_2^-$. Nitrobacter does $NO_2^- \to NO_3^-$.
Succession always starts from primary succession	WRONG. Secondary succession is more common and faster — occurs after disturbances (fire, farming abandonment) where soil exists.

UNIT 1 — MASTER SUMMARY TREE

UNIT 1 FUNDAMENTALS
■■■ 1. SCOPE & TERMS: Ecology (Haeckel,1866) | Ecosystem (Tansley,1935) | Biosphere (Suess,1875)
■■■ 2. EARTH SYSTEMS: Atmosphere (5 layers) | Hydrosphere | Lithosphere | Biosphere
■ ■■■ Troposphere (weather) → Stratosphere (ozone) → Mesosphere (meteors) → Thermosphere (auroras)
■■■ 3. ECOSYSTEM: Biotic (Producers, Consumers 1-3, Decomposers, Detritivores) + Abiotic (T, light, water, soil)
■ ■■■ Types: Terrestrial | Freshwater | Marine | Artificial
■■■ 4. ECOSYSTEM SERVICES: Provisioning | Regulating | Cultural | Supporting (PRCS)
■■■ 5. ENERGY FLOW: 10% law (Lindeman,1942) | Unidirectional | Laws of Thermodynamics
■ ■■■ GPP → NPP (= GPP – Ra) → NEP (= NPP – Rh)
■ ■■■ Food chains: Grazing (starts w/ plants) | Detritus (starts w/ dead matter)
■■■ 6. PYRAMIDS: Numbers (can invert) | Biomass (can invert) | Energy (NEVER inverts)
■■■ 7. BIOGEOCHEMICAL CYCLES:
■ ■■■ Carbon: Photosynthesis ↔ Respiration | Oceans = largest reservoir
■ ■■■ Nitrogen (most complex): Fix→Ammoni→Nitrif(Nitrosomonas→Nitrobacter)→Assim→Denitrif
■ ■■■ Phosphorus: SEDIMENTARY only (no gaseous phase) | Limiting nutrient in aquatic
■ ■■■ Water: Evaporation→Condensation→Precipitation→Infiltration→Runoff
■■■ 8. ABIOTIC FACTORS: Temperature (most important) | Light | Water | Soil | Wind | pH
■ ■■■ Laws: Liebig (minimum/limiting) | Shelford (tolerance min-opt-max)
■■■ 9. ECOLOGICAL CONCEPTS:
■ ■■■ Habitat (address) vs Niche (profession) | Fundamental vs Realized niche (Hutchinson)
■ ■■■ Gause's Competitive Exclusion Principle
■ ■■■ Succession: Primary (bare rock) vs Secondary (disturbed soil) → Climax community
■ ■■■ Population: Exponential (J-curve) | Logistic (S-curve,K = carrying cap.) | Max growth at K/2
■ ■■■ Interactions: Mutualism(++) | Commensalism(+0) | Amensalism(-0) | Predation(+-) | Parasitism(+-) | Competition(--)
■■■ 10. BIODIVERSITY: Alpha/Beta/Gamma | Hotspots (Myers,1988; 36 global; India=4: WISH)
■■■ Conservation: In-situ (NP, sanctuary) | Ex-situ (zoo, seed bank) | IUCN: EX-EW-CR-EN-VU-NT-LC

YOU ARE READY FOR UNIT 1 WHEN YOU CAN:

✓ Recall who coined ecology vs ecosystem and the exact years without hesitation ✓ Name all 5 atmosphere layers with their key features (weather, ozone, meteors, auroras) ✓ Calculate energy at any trophic level using 10% law ✓ State which pyramid is NEVER inverted and why ✓ Trace the complete nitrogen cycle with all organism names ✓ State in one sentence why phosphorus cycle is sedimentary ✓ Distinguish in-situ from ex-situ conservation with examples ✓ Name India's 4 biodiversity hotspots ✓ Explain difference between fundamental and realized niche ✓ State Liebig's and Shelford's laws without confusion Next units to study: Unit 3 (Environmental Biology) — HIGHEST weightage | Unit 5 (Pollution & Control) — MOST PYQs